

Time and Motion

Science — Physics · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer ($1 \times 5 = 5$)

1. State the SI unit of speed. [1]
2. Name the device used to measure the speed of a vehicle. [1]
3. Define the time period of a pendulum. [1]
4. Write the formula for speed. [1]
5. Name the type of motion shown by a swinging pendulum. [1]

Section B — Short Answer ($2 \times 5 = 10$)

1. A train covers 150 km in 3 hours. Calculate its speed. [2]
2. Convert 90 km/h into metres per second. [2]
3. Give two differences between uniform and non-uniform motion. [2]
4. What is a simple pendulum? Name the factor that decides its time period. [2]
5. How can the speed of a body be found from its distance–time graph? [2]

Section C — Long Answer ($5 \times 3 = 15$)

1. A body covers 60 m in the first 4 s, 40 m in the next 4 s and 80 m in the next 4 s. Is its motion uniform or non-uniform? Find the total distance, total time and average speed. [5]
2. Describe how a simple pendulum can be used to measure time, defining oscillation and time period. [5]
3. With an example of each, explain the distance–time graph of a body (i) at rest, (ii) in uniform motion and (iii) in non-uniform motion. [5]

Section D — Higher-Order Thinking (Give Reasons) ($2 \times 2 = 4$)

1. Two cars start from the same point in the same direction; car A at 40 km/h and car B at 60 km/h. How far apart are they after 2 hours? [2]
2. If the thread of a pendulum is shortened, will each oscillation take more or less time? Give a reason. [2]